

BENQODHI WORKSHOP

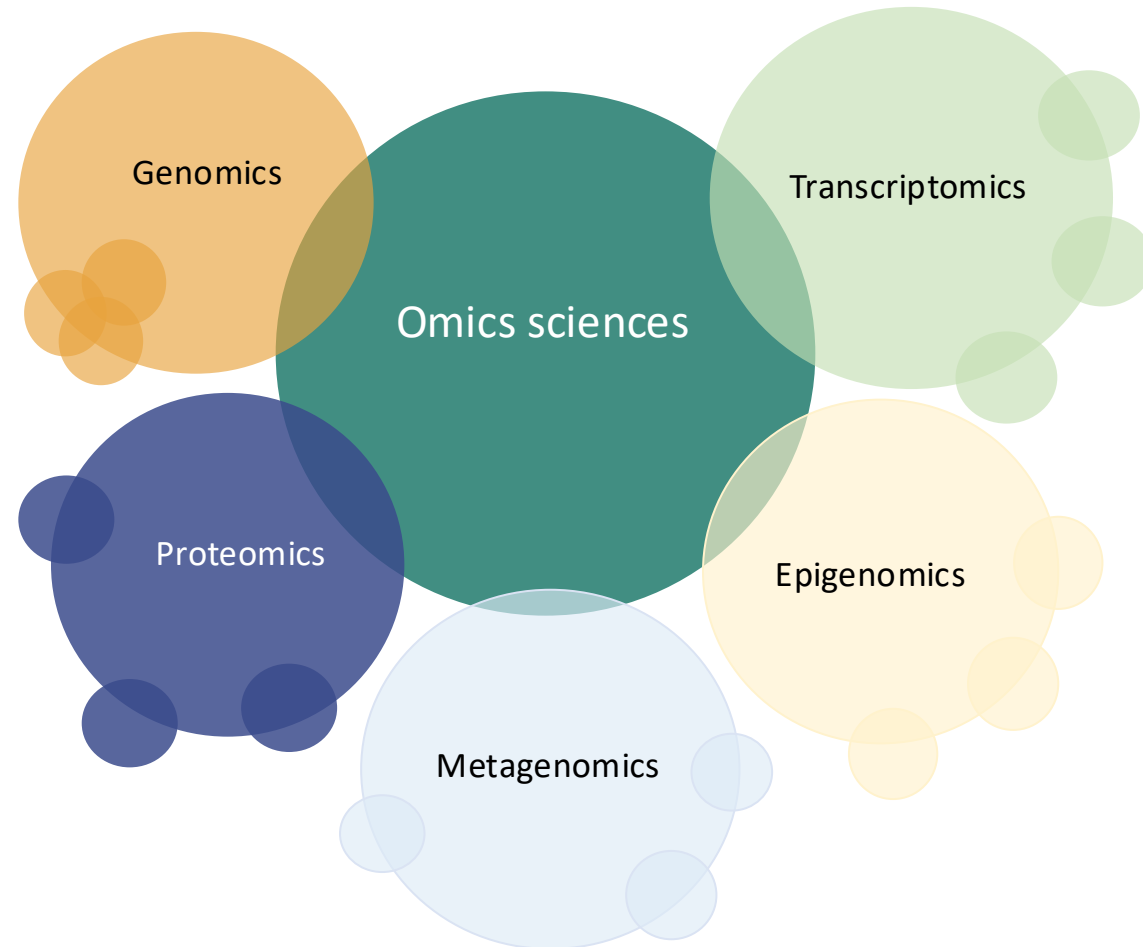
Where Genomics Resists Quantum Speedup : a biologist's POV

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Based on: Maurizio & Mazzola, "Quantum Computing for Genomics: Conceptual Challenges and Practical Perspectives," PRX Life 3, 047001 (2025)

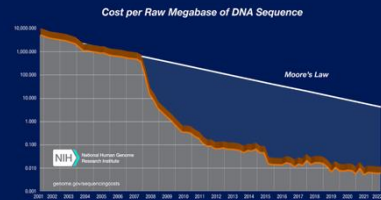
The omics universe



The promise vs. the premise

The promise

- Omics datasets scale into billions of base pairs and thousands of samples



- Synthetic datasets entered the game

Quantum algorithms promise speedups for search, optimization, and ML

The premise

But does the biology actually cooperate?

- Every candidate task competes against decades of tuned heuristics
- Speedup theorems rely on assumptions real biological data often violates
- “Combinatorially hard” in theory $\neq$ hard for today’s best classical solvers

Where does genomics actually open a door for quantum computing?

1

Not just “big” data

Genomic data is also noisy, and intricate— scale alone doesn’t make it quantum-friendly.

2

Combinatorial hardness ≠ quantum-friendly

Assembly, phasing, and tree inference are combinatorial — but that doesn’t automatically favor quantum methods.

3

The classical bar is high

Heuristics for these exact problems are already excellent and improving.

4

The real question

Where to start from?
The biological problem or from the quantum algorithm?

Quantum Computing for Genomics: Conceptual Challenges and Practical Perspectives

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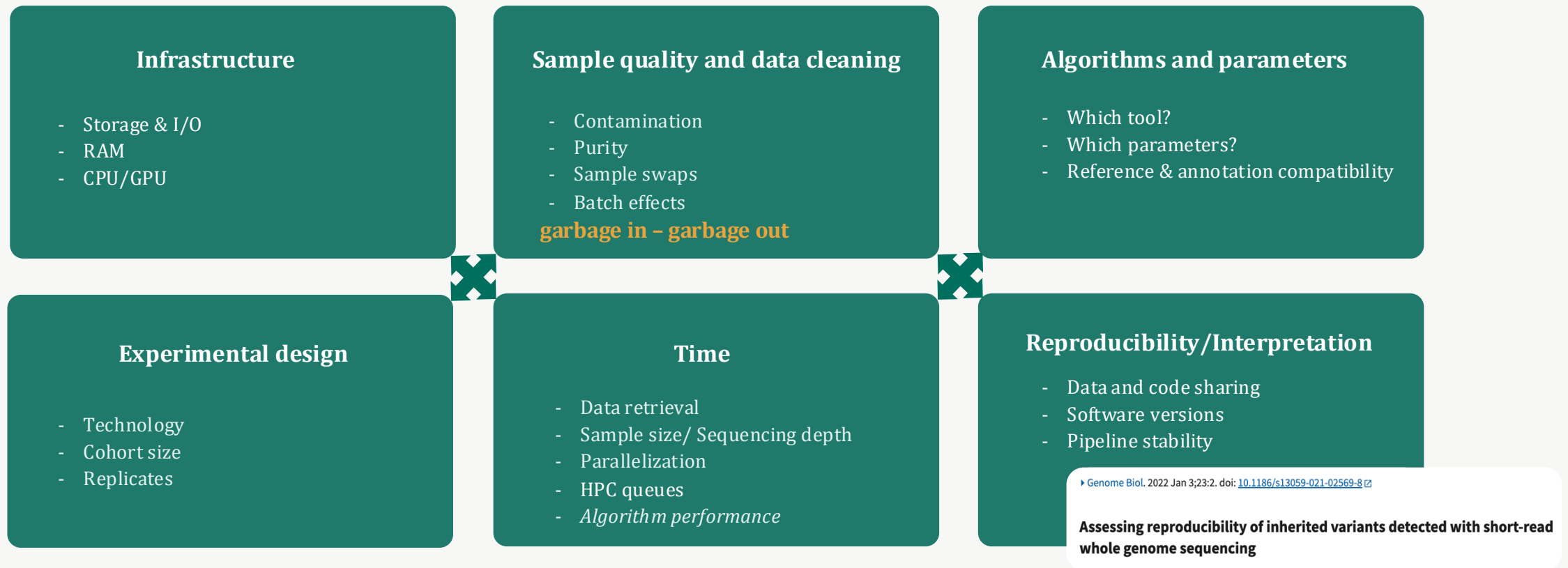
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Genomics 101, for a quantum audience

Not every computational bottleneck in genomics is a quantum-computing problem



Genomics 101, for a quantum audience

Processing Time

Task	Dataset description	Dataset size	Typical wall-clock time*	Main resource
RNA-seq DGE	10 samples	~0.01 GB (Expression matrix)	minutes	Laptop
scRNA-seq clustering	50k cells	~0.5 GB (Sparse count matrix + metadata)	minutes–hours	HPC cluster
Read alignment	40× human WGS	~50 GB (Fastq.gz)	hours	HPC cluster
Variant calling	40× human WGS	~50 GB (BAM) / ~30 GB (CRAM)	hours	HPC cluster
Phylogenetic tree	1000 taxa	~0.1-0.5 GB (alignment matrix)	hours–days	HPC cluster
De novo assembly	short reads 50x ~1 Gb genome	~50 GB (Fastq.gz)	hours–days	HPC cluster

*Very hardware-, parallelization-, coverage-, species, sample size, and implementation-dependent.

- Mapping sequencing reads to the reference genome is as simple as finding an *exact* string of length M within a database of length N.

Many problems in genomics defy conventional complexity classes:

- input data itself can contain experimental errors
- we may be interested in reasonably good solutions rather than the exact one

PER READS

SE READS

DNA FRAGMENT

LTR retrotransposons 6%

DNA transposons 3%

Simple sequence repeats 3%

Segmental duplications 5%

Misc. heterochromatin 6%

Misc. unique sequences 12%

SINEs 13%

LINEs 20%

Introns 20%

Protein coding genes 1.5%

Genomics 101, for a quantum audience

The reference genome

The promise

Complete genomes loaded in a quantum computer

Hepatitis D 1700 bp

The human virus with the smallest genome
Encoded in 117 qubits

First complete genome loaded onto a quantum computer

Researchers encode the tiny hepatitis D virus in an early step toward “quantum genomics”



The premise

Data loading could be an issue (memory and time overhead)

Genomes are measured in
base pairs (bp)

Human 3.2 Gb

We have around 3.2 billion base pairs in our genome.

Eucalyptus 640 Mb

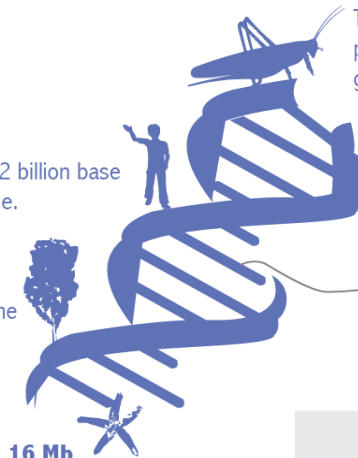
The Rose Gum eucalyptus genome has 640 million base pairs.

Starfish 16 Mb

The Asterina Pectinifera starfish has a 16 million base pair genome.

Grasshopper 18 Gb

The Mountain grasshopper, *Podisma pedestris*, has 18 billion base pairs in its genome.



A **base pair (bp)** is one
“rung” on the DNA ladder.

The base pair scale:

Kilo base pair (**Kb**) = 1,000 bp
Mega base pair (**Mb**) = 1 million base pairs
Giga base pair (**Gb**) = 1 billion base pairs

Database search meets real sequencing data

In theory

Grover's algorithm searches an unsorted database of N items in $\sqrt{N}$ steps — a quadratic speedup over classical linear search $O(N)$.

Tempting for: scanning read libraries, matching reference sequences, pattern search across genomes.

$O(\sqrt{N})$ speedup for searching an unstructured DB

In practice

- The number of qubits available on current quantum computers is limited
- Loading classical genomic data into quantum state (QRAM) costs at least as much as the search itself
- Realistic error rates and circuit depth erode the quadratic advantage on real quantum hardware
- Classical index structures (BLAST, hash tables, de Bruijn graphs) already search near-optimally

Net result: the expected speedup vanishes under realistic assumptions

What would a genuinely favorable biological problem look like?

Candidates that may fit:

De novo assembly

Gene / marker selection

Haplotype phasing

Phylogenetic tree

Clonal reconstruction

Limited variables

A relatively small, well-bounded number of decision variables — tractable for current or near-term hardware

Genuinely hard classically

A core optimization structure that resists today's best classical approximate solvers

Theoretical NP-hardness $\neq$ practical difficulty

Well-defined objective

Quantum optimization needs clean, quantifiable cost function — not a fuzzy or heuristic-only target

Machine learning in omics sciences

Omics applications

- Genetic variant calling
- Variant effect prediction
- Cell-type annotation
- Multimodal data integration

Benefits

- Fast prediction
- Handles high dimensionality
- Learns complex patterns
- Captures nonlinear relationships

Traditional limits

- Data complexity
- Data quality
- *Data availability*
- Overfitting
- Artifacts
- Reproducibility
- Generalizability
- Explainability
- Heavy computational scaling

Quantum ML limits

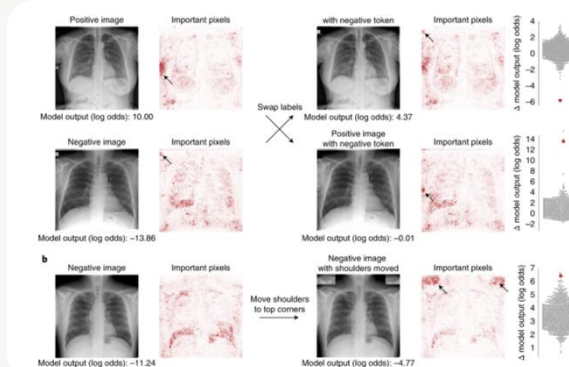
- Traditional limits of ML
- Traditional limits of QC
- Tests run on simulators
- Tests run on small/artificial/unimodal datasets
- Barren plateaus

What is PantheonOS?

PantheonOS is an evolvable, privacy-preserving multi-agent framework designed to reconcile generality with domain specificity. Autonomous agents powered by large language models collaborate to conduct biological discovery — achieving super-human performance on specialized scientific tasks.



The shared vision of the ELIXIR Federated Human Data Community is to operate a Europe-wide network of sensitive data archives that are able to function interoperably.



The saliency maps emphasize laterality tokens and shoulder position instead of lung opacity to differentiate between COVID-19-positive and -negative chest radiographs in the deep learning model.

nature > nature machine intelligence > articles > article

Article | Published: 31 May 2021

AI for radiographic COVID-19 detection selects shortcuts over signal

Alex J. DeGraaf, Joseph D. Janzani & Su-In Lee

Nature Machine Intelligence 3, 610–619 (2021) | [Cite this article](#)

LOOKING AHEAD

The bottleneck usually isn't the qubits — it's the biology.

Where quantum scientists and biologists need to collaborate

1

Problem selection & Shared vocabulary

Which real genomic instances are hard for classical solvers?
Which general classes of optimization problems amenable to quantum speedup appear in genomics?
A common language to correctly define the problems from the start, rooted in real biological constraints is needed.

2

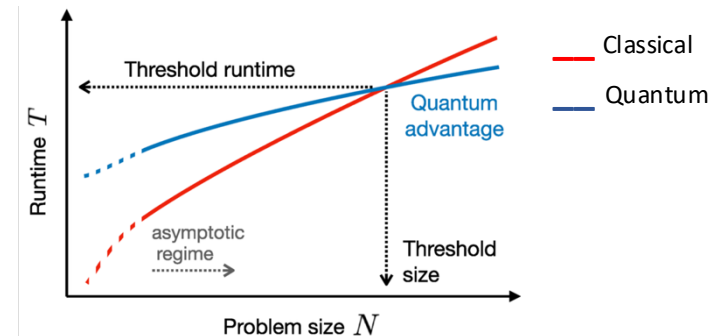
Honest benchmarking

Rigorous runtime-scaling studies against the best classical baselines, not naïve ones

3

Data sharing

Closer collaborations with the healthcare and biotech industries, which have access to multitudes of real-world data.





Thank you

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